Does Money Buy Wins? Payroll Efficiency and Competitive Performance in Major League Baseball (2021–2026)

Borna Anic-Matic

2026-10-30

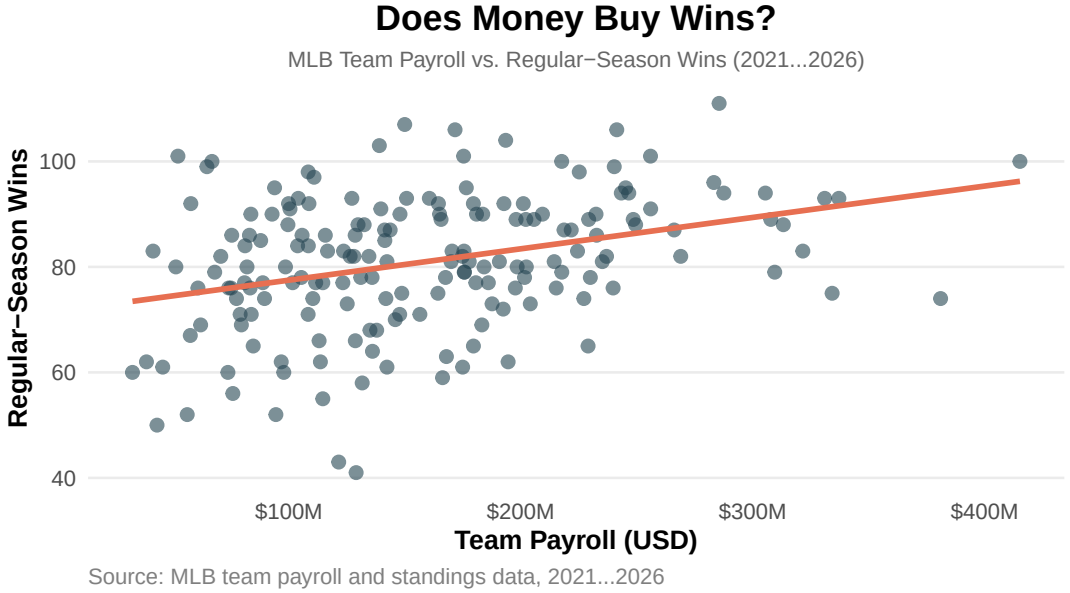
As the 2026 MLB season comes to an end, with the Milwaukee Brewers crowned World Series champions for the first time in franchise history, discussions surrounding a potential lockout and the introduction of salary caps and floors have raised an important question: **To what extent does a team's payroll predict its success?**

This project aims to investigate the relationship between team payroll and regular-season success in Major League Baseball from 2021 to 2026, examining whether payroll can serve as a reliable predictor of a team's performance.

Examining the Relationship Between Team Payroll and Regular-Season Performance (2021–2026)

To begin the analysis, we examine the relationship between team payroll and regular-season performance across Major League Baseball from 2021 to 2026. Specifically, we compare each team's annual payroll with its corresponding number of wins and losses.

Scatterplots are first used to visualize the direction and strength of these relationships. Pearson correlation coefficients are then calculated to quantify the linear associations between payroll and team performance, followed by coefficients of determination R^2 to evaluate the proportion of variation in wins and losses explained by payroll in a simple linear model.



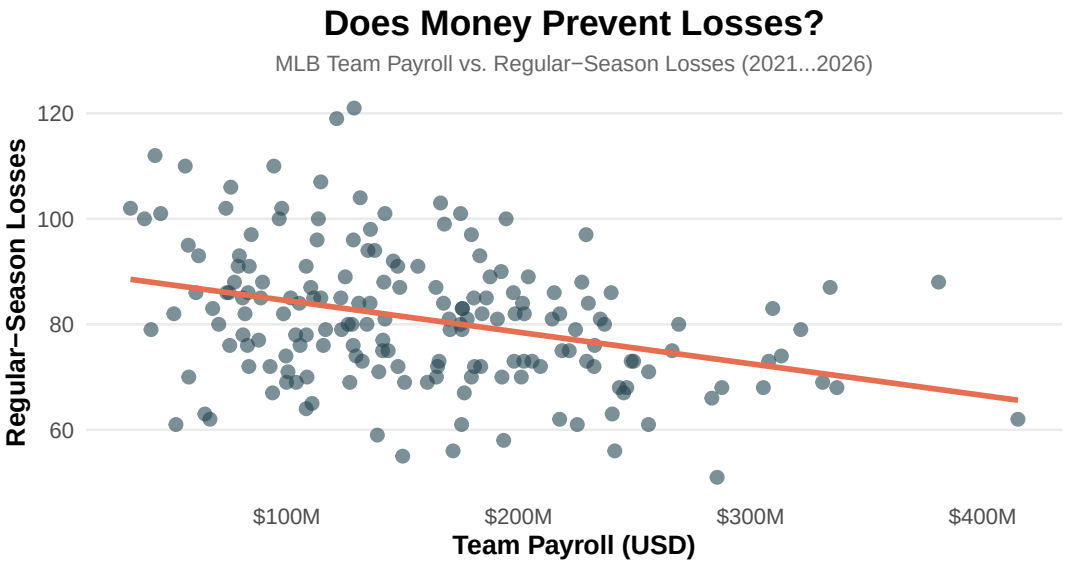


Table 1: Relationship Between MLB Team Payroll and Regular-Season Performance (2021–2026)

Relationship	Pearson's r	R^2
Payroll vs. Wins	0.3417	0.1168
Payroll vs. Losses	-0.3431	0.1177

The relationship between team payroll and regular-season wins is positive, with a Pearson correlation coefficient of 0.3417, indicating a weak-to-moderate linear association. An R-squared value of 0.1168 suggests that approximately 11.68% of the variation in team wins is explained by payroll in the simple linear model.

Conversely, the relationship between payroll and losses is negative, with a correlation coefficient of -0.3431. The corresponding R-squared value of 0.1177 indicates that approximately 11.77% of the variation in losses is explained by payroll.

Overall, these results suggest that while higher payrolls are generally associated with more wins and fewer losses, payroll alone explains a relatively small proportion of the variation in regular-season performance.

Examining the Relationship Between Year-Over-Year Changes in Team Payroll and Regular-Season Performance (2021–2026)

While the previous section examined the relationship between total team payroll and regular-season performance, this section investigates whether changes in payroll are associated with changes in team success from one season to the next.

To examine this relationship, we calculated the year-over-year differences in team payroll, wins, and losses. For each organization, the previous season's value was subtracted from the current season's value, allowing us to examine whether increases or decreases in spending corresponded with improvements or declines in regular-season performance.

$$\Delta P_{it} = P_{it} - P_{i,t-1}$$

$$\Delta W_{it} = W_{it} - W_{i,t-1}$$

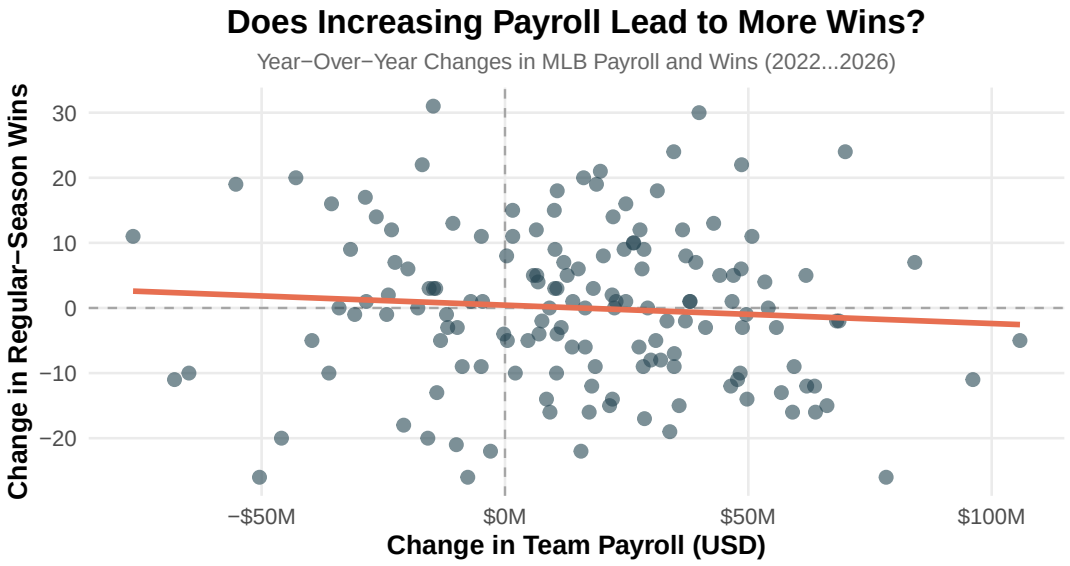
$$\Delta L_{it} = L_{it} - L_{i,t-1}$$

Because the dataset covers the 2021–2026 seasons, year-over-year comparisons were conducted from 2022 to 2026, resulting in 150 potential team-season comparisons.

Table 2: Average Year-Over-Year Changes in MLB Team Performance and Payroll (2022–2026)

Metric	Year-Over-Year Average Change
Average Change in Wins	0.00 wins
Average Change in Losses	0.00 losses
Average Change in Payroll	$15,371,866

Across the 2022–2026 period, the average year-over-year changes in regular-season wins and losses were both 0.00, while team payroll increased by an average of approximately 15.37 million per season. The absence of an average change in wins and losses is mathematically expected, as the total number of victories and defeats across MLB remains constant between complete regular seasons. Consequently, improvements in one team's record are offset by declines elsewhere in the league. However, what is particularly noteworthy is that despite this mathematical consistency in overall wins and losses, **MLB team payrolls increased by an average of 15.37 million annually**. This highlights a broader trend of increasing financial expenditure across the league, even as the total number of available regular-season victories remains effectively unchanged. Although this observation does not establish whether additional spending improves individual team performance, it raises an important question about the relationship between rising payrolls and competitive success, which is examined in the following correlation analysis.



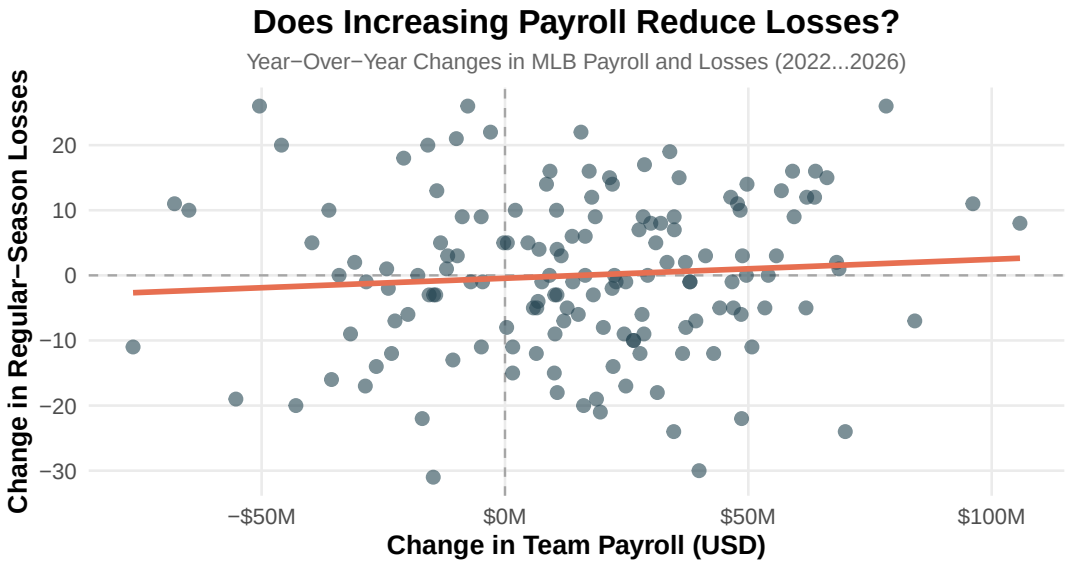


Table 3: Relationship Between Year-Over-Year Changes in MLB Payroll and Performance (2022–2026)

Relationship	Pearson's r	R^2
Payroll Change vs. Wins Change	-0.0778	0.0061
Payroll Change vs. Losses Change	0.0800	0.0064

The relationship between year-over-year changes in team payroll and changes in regular-season performance appears to be extremely weak.

The correlation between changes in payroll and wins was -0.0778, indicating a very weak negative linear relationship. An R-squared value of 0.0061 suggests that approximately 0.61% of the variation in year-over-year changes in wins is explained by changes in team payroll in the simple linear model.

Conversely, the correlation between changes in payroll and losses was 0.0800, indicating a very weak positive linear relationship. The corresponding R-squared value of 0.0064 indicates that approximately 0.64% of the variation in year-over-year changes in losses is explained by changes in payroll.

Overall, these findings suggest that increases or decreases in team payroll have little linear association with improvements or declines in regular-season performance. While changes in spending may influence individual teams, payroll changes alone are not reliable predictors of year-over-year changes in wins or losses across Major League Baseball.

Developing and Evaluating a Payroll Efficiency Metric in Major League Baseball

To further investigate the relationship between team payroll and regular-season performance, this section will introduce a measure of payroll efficiency. The objective is to determine whether MLB organizations are receiving sufficient value from their financial investments in player salaries.

Initial Measure: Cost per Win

To begin our analysis of payroll efficiency, we developed a simple cost-per-win metric that measures the amount of money an MLB organization spends in player payroll for each regular-season victory.

The metric is calculated as follows:

$$\text{Cost per Win}_{it} = \frac{P_{it}}{W_{it}}$$

Where:

P_{it} represents team $i's$ total payroll in season t.

W_{it} represents the number of regular-season wins achieved by team i in season t.

Under this calculation, a lower value indicates that a team spends fewer dollars per victory. However, this metric does not independently account for competitive success, meaning that teams with low payrolls can rank highly despite finishing with losing records.

Table 4: Top 5 MLB Teams by Lowest Cost per Win (2026)

Rank	Team	Year	Wins	Cost per Win
1	Miami Marlins	2026	80	$1,024,268
2	Cleveland Guardians	2026	85	$1,033,271
3	Tampa Bay Rays	2026	98	$1,105,365
4	Chicago White Sox	2026	84	$1,290,234
5	Milwaukee Brewers	2026	103	$1,349,518

The initial cost-per-win analysis revealed that the Milwaukee Brewers ranked among the five most financially efficient teams in 2026. With 103 regular-season wins, the Brewers spent approximately 1,349,518 dollars per victory. However, the Miami Marlins ranked first, spending approximately 1,024,268 dollars per win.

Miami's ranking highlights a limitation of using cost per win as the sole measure of payroll efficiency. Despite finishing with only 80 wins, two games below a .500 winning percentage, the Marlins were technically the most financially efficient team in baseball under this metric. This raises an important question: should a team be considered financially efficient simply because it spends relatively little, even if it fails to achieve a winning record?

Revised Measure: Payroll Efficiency Index

To address the limitations of the initial cost-per-win metric, we developed a revised efficiency index that incorporates team performance relative to the league average.

First, we calculated the average number of regular-season wins across all MLB team-season observations from 2021 to 2026:

$$\overline{W} = \frac{\sum_{i,t} W_{it}}{N}$$

Where N represents the total number of team-season observations in the dataset.

Using this baseline, we constructed the following payroll efficiency index:

$$E_{it} = \frac{W_{it} - \overline{W}}{P_{it}/1{,}000{,}000}$$

Where:

- E_{it} represents the payroll efficiency score for team i in season t.
- W_{it} represents team i's regular-season wins in season t.
- $\overline{W}$ represents the average number of regular-season wins across all MLB teams from 2021 to 2026.
- P_{it} represents team i's total payroll in US dollars for season t.

- i represents the individual MLB team.
- t represents the season being analyzed.

This index measures the number of wins above league average achieved per $1 million in team payroll. A positive score indicates above-average regular-season performance, while a negative score indicates below-average performance. Higher scores represent greater payroll-adjusted success under this metric.

Unlike the original cost-per-win calculation, the revised index distinguishes teams that achieve winning results from those that spend relatively little but perform below league average.

Evaluating Long-Term Payroll Efficiency

While individual-season efficiency scores provide insight into annual performance, they do not necessarily reflect an organization's ability to sustain payroll efficiency over time.

To investigate long-term performance, we calculated the arithmetic mean of each team's annual efficiency scores from 2021 to 2026:

$$\overline{E}_i = \frac{1}{6} \sum_{t=2021}^{2026} E_{it}$$

Where $\overline{E}_i$ represents team $i's$ average annual efficiency score over the six seasons.

This calculation assigns equal weight to each season, allowing us to compare organizations based on their average payroll-adjusted performance rather than relying on a single exceptional year.

However, because this approach averages annual ratios, it does not necessarily produce the same rankings as dividing total wins above average by total payroll across the six seasons. This distinction will be examined in the subsequent sensitivity analysis.

Table 5: Top 5 MLB Teams by Payroll Efficiency, 2021–2026

Year	Rank	Team	Wins	Efficiency Score
2021	1	Tampa Bay Rays	100	0.2846
2021	2	San Francisco Giants	107	0.1735
2021	3	Milwaukee Brewers	95	0.1494
2021	4	Seattle Mariners	90	0.1079
2021	5	Los Angeles Dodgers	106	0.1036
2022	1	Cleveland Guardians	92	0.1909
2022	2	Houston Astros	106	0.1458
2022	3	Atlanta Braves	101	0.1142
2022	4	Los Angeles Dodgers	111	0.1051
2022	5	Seattle Mariners	90	0.0972
2023	1	Baltimore Orioles	101	0.3838
2023	2	Tampa Bay Rays	99	0.2787
2023	3	Atlanta Braves	104	0.1190
2023	4	Milwaukee Brewers	92	0.1014
2023	5	Los Angeles Dodgers	100	0.0874
2024	1	Milwaukee Brewers	93	0.1155
2024	2	Cleveland Guardians	92	0.1103
2024	3	Baltimore Orioles	91	0.0997
2024	4	Los Angeles Dodgers	98	0.0756
2024	5	San Diego Padres	93	0.0748
2025	1	Milwaukee Brewers	97	0.1445
2025	2	Cleveland Guardians	88	0.0705
2025	3	Seattle Mariners	90	0.0610
2025	4	Chicago Cubs	92	0.0572
2025	5	Toronto Blue Jays	94	0.0535
2026	1	Milwaukee Brewers	103	0.1584
2026	2	Tampa Bay Rays	98	0.1571
2026	3	Atlanta Braves	94	0.0528
2026	4	Los Angeles Dodgers	100	0.0458
2026	5	Cleveland Guardians	85	0.0457

Examining the annual payroll efficiency rankings from 2021 to 2026 reveals several interesting patterns. Most notably, the Milwaukee Brewers ranked among the five most efficient teams in five of the six seasons analyzed, suggesting a consistent ability to achieve above-average regular-season success relative to their payroll. Similarly, the Tampa Bay Rays frequently appeared near the top of the rankings, indicating that they have also consistently performed well relative to their financial resources.

Another notable observation is the consistent presence of the Los Angeles Dodgers among the five most efficient teams. Unlike the Brewers and Rays, the Dodgers have operated with substantially larger payrolls, frequently ranking among MLB's highest-spending organizations. Their strong efficiency rankings demonstrate that high spending does not necessarily prevent a team from achieving favourable payroll-adjusted

results. This contrasts with the New York Mets, another organization known for its substantial payroll, which did not appear among the annual top-five rankings presented. This difference suggests that financial resources alone do not guarantee efficient regular-season performance.

Table 6: Top 10 MLB Teams by Average Payroll Efficiency (2021–2026)

Rank	Team	Average Efficiency Score
1	Tampa Bay Rays	0.1220
2	Milwaukee Brewers	0.1180
3	Los Angeles Dodgers	0.0756
4	Atlanta Braves	0.0585
5	Cleveland Guardians	0.0553
6	Houston Astros	0.0551
7	Seattle Mariners	0.0539
8	New York Yankees	0.0428
9	Philadelphia Phillies	0.0341
10	Toronto Blue Jays	0.0323

When comparing average payroll efficiency scores across the entire 2021–2026 period, the Tampa Bay Rays and Milwaukee Brewers emerge as the two highest-ranked organizations, with average scores of 0.1220 and 0.1180, respectively. A noticeable decline follows, with the Los Angeles Dodgers ranking third (0.0756), followed by the Atlanta Braves (0.0585) and Cleveland Guardians (0.0553). Notably, the Guardians' average efficiency score reflects their complete six-season history, including the 2021 season when the franchise operated as the Cleveland Indians.

Overall, these findings suggest that certain MLB organizations have demonstrated a sustained ability to achieve above-average regular-season results relative to their payroll. While the Rays and Brewers stand out for their consistently strong efficiency scores, the Dodgers demonstrate that organizations with substantial financial resources can also perform favourably under this metric. However, these rankings remain dependent on the definition of payroll efficiency used and should not be interpreted as definitive measures of organizational effectiveness.

Sensitivity Analysis and Validation of the Payroll Efficiency Index

The previous section introduced a payroll efficiency index designed to measure regular-season wins above the leaguewide average relative to team payroll. Although this metric provides a method for comparing organizational performance across Major League Baseball, its results depend on several methodological decisions, particularly the selection of a performance baseline and the approach used to aggregate efficiency across multiple seasons.

To evaluate the robustness of the proposed index, two sensitivity analyses were conducted. The first investigated how alternative regular-season win baselines influenced organizational efficiency scores and rankings. The second examined whether rankings changed when long-term efficiency was calculated using an aggregate measure rather than the arithmetic mean of annual efficiency scores.

The purpose of these analyses was not to identify a universally correct measure of payroll efficiency, but rather to determine the extent to which the conclusions drawn from the original index remained consistent under alternative methodological assumptions.

Sensitivity to Alternative Win Baselines

Table 7: Sensitivity of MLB Payroll Efficiency Scores to Different Win Baselines (2021–2026)

Rank	Team	70 Wins	75 Wins	81 Wins	85 Wins	90 Wins
1	Tampa Bay Rays	0.2581	0.1962	0.1218	0.0722	0.0103
2	Milwaukee Brewers	0.2159	0.1714	0.1179	0.0822	0.0377
3	Los Angeles Dodgers	0.1161	0.0977	0.0756	0.0608	0.0424
4	Atlanta Braves	0.1161	0.0898	0.0584	0.0374	0.0111
5	Cleveland Guardians	0.2047	0.1367	0.0551	0.0007	-0.0673
6	Houston Astros	0.1098	0.0849	0.0550	0.0351	0.0103
7	Seattle Mariners	0.1442	0.1031	0.0537	0.0208	-0.0203
8	New York Yankees	0.0842	0.0654	0.0428	0.0277	0.0089
9	Philadelphia Phillies	0.0804	0.0593	0.0340	0.0171	-0.0040
10	Toronto Blue Jays	0.0870	0.0621	0.0323	0.0124	-0.0125
11	San Diego Padres	0.0861	0.0613	0.0315	0.0117	-0.0131
12	Boston Red Sox	0.0720	0.0462	0.0152	-0.0054	-0.0312
13	San Francisco Giants	0.0781	0.0492	0.0145	-0.0086	-0.0375
14	New York Mets	0.0477	0.0300	0.0088	-0.0054	-0.0231
15	St. Louis Cardinals	0.0794	0.0448	0.0033	-0.0243	-0.0589
16	Chicago Cubs	0.0599	0.0324	-0.0006	-0.0225	-0.0500
17	Baltimore Orioles	0.1447	0.0766	-0.0051	-0.0596	-0.1278
18	Detroit Tigers	0.0710	0.0292	-0.0211	-0.0545	-0.0964
19	Minnesota Twins	0.0593	0.0214	-0.0241	-0.0544	-0.0923
20	Cincinnati Reds	0.0698	0.0229	-0.0333	-0.0707	-0.1175
21	Texas Rangers	0.0212	-0.0096	-0.0467	-0.0714	-0.1022
22	Arizona Diamondbacks	0.0390	-0.0018	-0.0507	-0.0833	-0.1241
23	Los Angeles Angels	-0.0005	-0.0272	-0.0593	-0.0806	-0.1074
24	Chicago White Sox	-0.0074	-0.0485	-0.0979	-0.1308	-0.1719
25	Miami Marlins	0.0466	-0.0197	-0.0993	-0.1523	-0.2186
26	Kansas City Royals	0.0048	-0.0437	-0.1019	-0.1407	-0.1892
27	Washington Nationals	-0.0185	-0.0633	-0.1171	-0.1529	-0.1977
28	Colorado Rockies	-0.0702	-0.1078	-0.1530	-0.1832	-0.2208
29	Pittsburgh Pirates	-0.0221	-0.1019	-0.1976	-0.2614	-0.3411
30	Athletics	-0.0929	-0.1787	-0.2816	-0.3502	-0.4360

The original payroll efficiency index measured regular-season wins above the leaguewide average per $1 million in payroll. However, selecting league-average wins as the benchmark represents a methodological decision that may influence how organizations are evaluated.

To investigate this limitation, five alternative win baselines were selected: 70, 75, 81, 85, and 90 wins. These values represent progressively more demanding performance thresholds, ranging from a relatively modest level of regular-season success to a benchmark more closely associated with competitive teams.

For each baseline, the efficiency index was recalculated for all 30 MLB organizations across the 2021–2026 period. The resulting annual scores were then averaged by franchise to produce separate six-year efficiency rankings under each assumption.

The revised index was calculated as follows:

$$E_{it}(B) = \frac{W_{it} - B}{P_{it}/1{,}000{,}000}$$

Where:

$E_{it}(B)$ represents the efficiency score for team (i) in season (t), using baseline (B).

W_{it} represents the team's regular-season wins.

B represents the selected win baseline.

P_{it} represents the team's payroll in US dollars.

The results demonstrated that the relative efficiency rankings were sensitive to the selected baseline. The Tampa Bay Rays ranked first under the 70-, 75-, and 81-win thresholds, while the Milwaukee Brewers overtook Tampa Bay at 85 wins. At the 90-win threshold, the Los Angeles Dodgers emerged as the highest-ranked organization.

These changes illustrate an important characteristic of the index. Because payroll appears in the denominator, the effect of increasing the baseline is greater for lower-payroll organizations. Consequently, teams with relatively modest payrolls may receive particularly favourable efficiency scores under less demanding thresholds but experience larger reductions as the baseline increases.

For example, Cleveland's ranking declined from third under the 70-win baseline to sixteenth under the 90-win baseline. Similarly, the Baltimore Orioles fell from fourth to twenty-third. In contrast, the Los Angeles Dodgers improved from sixth to first across the same range.

These findings suggest that the proposed index does not measure financial efficiency independently of competitive expectations. Rather, it evaluates the extent to which organizations exceed a specified performance benchmark relative to their financial expenditure. Therefore, the interpretation of efficiency depends substantially on how successful performance is defined.

Table 8: MLB Payroll Efficiency Rankings Across Alternative Win Baselines (2021–2026)

Overall	Team	70 Wins	75 Wins	81 Wins	85 Wins	90 Wins	Average Rank
1	Tampa Bay Rays	1	1	1	2	4	1.8
1	Milwaukee Brewers	2	2	2	1	2	1.8
3	Los Angeles Dodgers	6	5	3	3	1	3.6
4	Atlanta Braves	7	6	4	4	3	4.8
5	Houston Astros	8	7	6	5	5	6.2
6	Seattle Mariners	5	4	7	7	10	6.6
7	Cleveland Guardians	3	3	5	11	16	7.6
8	New York Yankees	11	9	8	6	6	8.0
9	Toronto Blue Jays	9	10	10	9	8	9.2
10	Philadelphia Phillies	12	12	9	8	7	9.6
11	San Diego Padres	10	11	11	10	9	10.2
12	Boston Red Sox	15	14	12	12	12	13.0
13	San Francisco Giants	14	13	13	14	13	13.4
14	Baltimore Orioles	4	8	17	19	23	14.2
15	St. Louis Cardinals	13	15	15	16	15	14.8
16	New York Mets	20	17	14	13	11	15.0
17	Chicago Cubs	18	16	16	15	14	15.8
18	Detroit Tigers	16	18	18	18	18	17.6
19	Minnesota Twins	19	20	19	17	17	18.4
20	Cincinnati Reds	17	19	20	20	21	19.4
21	Texas Rangers	23	22	21	21	19	21.2
22	Arizona Diamondbacks	22	21	22	23	22	22.0
23	Los Angeles Angels	25	24	23	22	20	22.8
24	Miami Marlins	21	23	25	26	27	24.4
25	Chicago White Sox	26	26	24	24	24	24.8
26	Kansas City Royals	24	25	26	25	25	25.0
27	Washington Nationals	27	27	27	27	26	26.8
28	Colorado Rockies	29	29	28	28	28	28.4
29	Pittsburgh Pirates	28	28	29	29	29	28.6
30	Athletics	30	30	30	30	30	30.0

While the initial sensitivity analysis demonstrated that individual rankings changed under different win baselines, it did not directly identify which organizations performed most consistently across the five scenarios.

To address this, a supplementary rank-based comparison was developed. Each organization was assigned a ranking from first to thirtieth under each baseline, with first place representing the highest average efficiency score. The arithmetic mean of these five rankings was then calculated to determine each organization's average placement.

The average-ranking measure was defined as:

$$\overline{R}i = \frac{1}{5} \sum B \in 70, 75, 81, 85, 90 R_{i,B}$$

Where:

- $\overline{R}_i$ represents the average ranking position of organization (i).
- $R_{i,B}$ represents the organization's ranking under baseline (B).

Lower average-ranking values indicate stronger overall placement across the selected baselines. The results revealed that the Tampa Bay Rays and Milwaukee Brewers tied for first place, each recording an average ranking of 1.8. However, their individual ranking patterns differed.

Tampa Bay ranked first under three baselines, second under the 85-win baseline, and fourth under the 90-win baseline. Milwaukee, conversely, ranked second under four baselines and first under the 85-win baseline. Although both organizations achieved the same average placement, Milwaukee demonstrated greater consistency by remaining within the top two under every scenario.

The Los Angeles Dodgers ranked third overall, with an average placement of 3.6, followed by the Atlanta Braves (4.8) and Houston Astros (6.2). These results suggest that certain organizations consistently performed well relative to their payroll even when the selected benchmark changed.

Other franchises displayed substantially greater ranking variability. The Cleveland Guardians recorded an average placement of 7.6 despite ranking third under the two lowest baselines. The Baltimore Orioles, meanwhile, recorded an average placement of 14.2, reflecting their considerable decline as the performance threshold increased.

Overall, the average-ranking approach provides an additional perspective on the robustness of the efficiency index by distinguishing organizations that consistently achieve favourable rankings from those whose positions depend more heavily on the selected baseline. However, this approach should be understood as a supplementary comparison rather than an independent measure of financial efficiency. The selected baselines are alternative methodological assumptions, and assigning equal weight to each does not establish that they are equally appropriate measures of competitive success.

Sensitivity to Alternative Aggregation Method

The preceding analyses demonstrated that changes in the performance baseline could substantially influence organizational efficiency rankings. However, another methodological consideration concerns how efficiency scores are combined across multiple seasons.

The original six-year analysis calculated the arithmetic mean of each organization's annual efficiency scores, assigning equal weight to every season. While this approach provides a straightforward measure of long-term efficiency, it does not account for variation in the amount of payroll expenditure between seasons.

To investigate whether this decision influenced the results, an alternative aggregate efficiency measure was calculated. Rather than averaging annual efficiency ratios, the aggregate approach divided each organization's total wins above the original leaguewide baseline by its combined payroll expenditure across all six seasons.

The aggregate efficiency measure was calculated as follows:

$$E_i^{\text{aggregate}} = \frac{\sum_{t=2021}^{2026}(W_{it} - \overline{W})}{\sum_{t=2021}^{2026} P_{it}/1{,}000{,}000}$$

Where:

$E_i^{\text{aggregate}}$ represents the aggregate efficiency score for organization i.

W_{it} represents regular-season wins in season t.

$\overline{W}$ represents the original leaguewide average number of wins.

P_{it} represents team payroll in season t.

Unlike the arithmetic mean of annual efficiency scores, this approach effectively weights each season's efficiency by its share of the organization's total payroll expenditure. Consequently, seasons with higher payrolls exert greater influence on the aggregate measure.

Table 9: Comparison of MLB Payroll Efficiency Rankings Under Alternative Aggregation Methods (2021-2026)

Team	Annual Average Rank	Aggregate Rank	Rank Change
Milwaukee Brewers	2	1	1
Tampa Bay Rays	1	2	-1
Los Angeles Dodgers	3	3	0
Cleveland Guardians	5	4	1
Atlanta Braves	4	5	-1
Houston Astros	6	6	0
Seattle Mariners	7	7	0
New York Yankees	8	8	0
Philadelphia Phillies	9	9	0
San Diego Padres	11	10	1
Toronto Blue Jays	10	11	-1
Boston Red Sox	12	12	0
New York Mets	14	13	1
St. Louis Cardinals	15	14	1
San Francisco Giants	13	15	-2
Chicago Cubs	16	16	0
Baltimore Orioles	17	17	0
Detroit Tigers	18	18	0
Minnesota Twins	19	19	0
Arizona Diamondbacks	22	20	2
Texas Rangers	21	21	0
Cincinnati Reds	20	22	-2
Los Angeles Angels	23	23	0
Kansas City Royals	26	24	2
Chicago White Sox	24	25	-1
Miami Marlins	25	26	-1
Washington Nationals	27	27	0
Pittsburgh Pirates	29	28	1
Colorado Rockies	28	29	-1
Athletics	30	30	0

The comparison revealed relatively minor changes in organizational rankings. Under the original annual-average approach, the Tampa Bay Rays ranked first with an efficiency score of 0.1220, followed by the

Milwaukee Brewers at 0.1180. Under the aggregate approach, however, Milwaukee overtook Tampa Bay, recording a score of 0.1169 compared with Tampa Bay's 0.1077.

The Los Angeles Dodgers retained third place under both approaches, while the Cleveland Guardians and Atlanta Braves exchanged fourth and fifth positions.

Across all 30 organizations, fourteen retained identical ranking positions, and no franchise moved more than two places. The ten highest-ranked organizations were also largely consistent, with the San Diego Padres replacing the Toronto Blue Jays in the aggregate top ten.

These findings suggest that, although the aggregation method can influence individual ranking positions, the overall ordering of organizations remains relatively stable.

Table 10: Spearman's Rank Correlation Between Alternative Payroll Efficiency Rankings (2021–2026)

Comparison	Spearman's rho	Teams Compared
Annual Average vs. Aggregate Efficiency Rankings	0.9938	30

To quantify the similarity between the two aggregation methods, Spearman's rank correlation coefficient was calculated.

Unlike Pearson's correlation, which measures the strength and direction of a linear relationship between numerical variables, Spearman's correlation evaluates the strength and direction of a monotonic relationship using ranked observations. In this analysis, it was used to determine how closely the organizational rankings generated by the two aggregation methods corresponded.

Spearman's rank correlation coefficient was calculated using the ranking positions of all 30 MLB organizations under the annual-average and aggregate approaches.

This indicates an extremely strong positive association between the two sets of rankings. Although individual organizations occasionally exchanged positions, the near-perfect correlation demonstrates that the overall ordering remained highly consistent across both methods.

The results therefore provide evidence that the original efficiency rankings are relatively robust to the choice between averaging annual efficiency scores and aggregating six-year payroll and performance.

However, a high rank correlation does not establish that either approach provides an objectively correct measure of organizational efficiency. Rather, it demonstrates that the two methods produce highly similar relative rankings within the dataset.

Discussion of Sensitivity Analysis Findings

Taken together, the two sensitivity analyses reveal an important distinction in the robustness of the proposed payroll efficiency index.

The selection of the regular-season win baseline had a substantial influence on the efficiency rankings of several organizations. In particular, teams with lower payrolls experienced greater changes in their efficiency scores as the baseline increased, while organizations with consistently high win totals, such as the Los Angeles Dodgers, became more favourably ranked under demanding performance thresholds.

By contrast, changing the method used to aggregate efficiency across six seasons produced relatively minor ranking differences. Spearman's rank correlation coefficient of 0.9938 demonstrated extremely strong agreement between the annual-average and aggregate approaches, suggesting that the broad organizational rankings were not heavily dependent on this particular methodological decision.

The Milwaukee Brewers and Tampa Bay Rays emerged as particularly noteworthy organizations throughout these analyses. Both consistently ranked near the top under alternative win baselines, tied for first in average

placement, and occupied the top two positions under both aggregation methods. Their results suggest that their favourable payroll-adjusted performance is relatively robust across the specific assumptions examined.

Nevertheless, these findings also reinforce the limitations of the index. Payroll efficiency is not a fixed organizational characteristic that can be measured independently of analytical choices. Rather, its evaluation depends on the definition of successful performance, the treatment of financial expenditure, and the manner in which results are summarized over time.

The sensitivity analyses therefore support using the proposed index as a comparative analytical tool while emphasizing the importance of clearly specifying its assumptions. Within the alternatives examined, the efficiency rankings were considerably more sensitive to the selection of the performance baseline than to the choice of aggregation method.

These findings provide a more informed foundation for the subsequent investigation of hypothetical salary caps and payroll floors, where differences in organizational spending will be examined under alternative financial structures.

Simulating the Financial and Economic Implications of MLB Salary Caps and Payroll Floors

Following the examination of the relationship between team payroll and regular-season performance, as well as the development and evaluation of a payroll efficiency index, this section shifts its focus toward the potential economic implications of introducing salary caps and payroll floors in Major League Baseball. While the previous analyses investigated how effectively organizations have utilized their financial resources, an equally important consideration is how leaguewide financial regulations might influence the distribution of those resources.

With discussions surrounding MLB's collective bargaining agreement and the potential introduction of a salary cap-and-floor system, questions regarding competitive balance, financial inequality, and player compensation have become increasingly relevant. MLB's reported November 2026 proposal, which outlined a 258.2 million dollar salary cap and a $180.2 million dollar payroll floor, provides a reference point for investigating the potential consequences of such regulations.

Using team payroll data from the 2026 season, this section conducts a series of financial simulations to examine how alternative spending restrictions could affect MLB organizations. Three scenarios are considered: one based on MLB's reported proposal and two hypothetical alternatives representing more restrictive and flexible financial structures. For each scenario, the analysis identifies organizations operating outside the proposed spending boundaries and calculates the financial adjustments required to achieve compliance.

Beyond evaluating changes in total leaguewide payroll expenditure, the analysis also introduces the Gini coefficient to measure how each scenario influences payroll inequality. This distinction is particularly important, as reducing disparities in organizational spending does not necessarily imply increasing total player compensation or improving competitive balance.

The objective of this section is therefore to evaluate the financial implications and potential trade-offs associated with alternative salary cap-and-floor structures. Because the simulations apply proposed thresholds to historical payroll data and assume immediate compliance, the results should be interpreted as illustrative economic scenarios rather than predictions of organizational behaviour or future competitive outcomes.

Evaluating the Financial Impact of MLB's Proposed Salary Cap and Payroll Floor

Table 11: Illustrative Financial Impact of MLB's Proposed 258.2M Salary Cap and 180.2M Payroll Floor Using 2026 Payroll Data

Payroll Status	Number of Teams	Total Required Adjustment
Above Cap	6	-$472.94M
Within Range	12	$0.00M
Below Floor	12	$736.67M
Total	**30**	**$263.72M**

Under the scenario based on MLB's November 2026 salary cap-and-floor proposal, six organizations exceeded the proposed 258.2 million dollar salary cap, while twelve fell below the $180.2 million payroll floor. The remaining twelve organizations operated within the specified spending range.

Based on the payroll figures used in this analysis, organizations exceeding the salary cap would collectively need to reduce expenditures by approximately 472.94 million dollars, while those below the payroll floor would need to increase spending by approximately 736.67 million dollars. These adjustments would result in a net increase of approximately 263.72 million dollars in leaguewide payroll expenditure, with a combined absolute adjustment of approximately 1.21 billion dollars.

Notably, the required payroll increases exceed the reductions, suggesting that the proposed financial structure could increase overall player spending while simultaneously narrowing the permitted range of team payrolls.

However, these results differ from MLB's published estimates, which identified eight organizations exceeding the cap and twelve below the floor, with a projected net spending increase of $108 million. This discrepancy highlights the importance of payroll definitions, particularly the distinction between the payroll figures used in this analysis and MLB's Competitive Balance Tax calculations.

Therefore, the simulation should be interpreted as an illustrative assessment of the proposed financial thresholds rather than an exact reproduction of MLB's projections. Furthermore, the calculations assume immediate compliance and do not incorporate transitional provisions, contractual obligations, or potential organizational responses.

Comparative Analysis of Alternative MLB Salary Cap and Payroll Floor Structures

Table 12: Financial Implications of Alternative MLB Salary Cap and Payroll Floor Scenarios (2026)

Scenario	Above Cap	Below Floor	Within Range	Reductions	Increases	Net Change	Absolute Change
A - MLB Proposal	6	12	12	$472.94M	$736.67M	$263.72M	$1209.61M
B - Restrictive	15	12	3	$1129.83M	$374.27M	-$755.57M	$1504.10M
C - Flexible	5	7	18	$256.04M	$139.08M	-$116.97M	$395.12M

Comparing Alternative Salary Cap and Payroll Floor Scenarios

To investigate how the financial implications of salary regulations depend on their structure, three alternative cap-and-floor scenarios were applied to the 2026 MLB payroll dataset. The first used the thresholds from MLB's reported November 2026 proposal, while the second and third represented hypothetical restrictive and flexible alternatives.

For each scenario, the number of organizations exceeding the salary cap, falling below the payroll floor, or operating within the permitted range was calculated. The minimum payroll adjustments required for compliance were then used to estimate the total increases, reductions, net spending changes, and absolute financial adjustments across the league.

The results revealed substantial differences between the three scenarios. Under Scenario A, based on MLB's proposed 258.2 million dollar cap and 180.2 million dollar floor, eighteen organizations would require payroll

adjustments. Required spending increases of approximately 736.67 million dollars exceeded reductions of 472.94 million dollars, producing a net leaguewide payroll increase of approximately 263.72 million dollars.

In contrast, Scenario B, which introduced a 200 million dollar cap and 150 million dollar floor, produced the greatest financial adjustments. Twenty-seven organizations would fall outside the permitted spending range, with fifteen exceeding the cap and twelve falling below the floor. Required payroll reductions of approximately 1.13 billion dollars substantially exceeded spending increases of 374.27 million dollars, resulting in a net reduction of approximately 755.57 million dollars. The total absolute adjustment of $1.50 billion was also the largest among the three scenarios.

Scenario C, which established a 300 million dollar cap and 125 million dollar floor, produced the smallest financial adjustments. Only twelve organizations would fall outside the permitted range, with a combined absolute adjustment of approximately 395.12 million dollars. Nevertheless, required reductions of 256.04 million dollars exceeded increases of 139.08 million dollars, producing a net payroll decrease of approximately $116.97 million.

These findings demonstrate that the introduction of a salary cap and payroll floor does not necessarily result in either increased or decreased leaguewide player spending. Rather, the direction and magnitude of the financial effects depend on the specific thresholds selected and the existing distribution of team payrolls.

Importantly, the scenarios also demonstrate that the number of organizations affected and the total financial adjustments required are distinct considerations. Although Scenario B affects the greatest number of organizations, its financial implications also differ substantially from those of Scenario A, which produces an increase rather than a decrease in total payroll expenditure.

However, these calculations assume that organizations adjust their payrolls only to the nearest required boundary. They do not account for contractual obligations, implementation periods, changes in organizational spending decisions, or the potential redistribution of financial resources across the league. Consequently, the results should be interpreted as simplified financial scenarios rather than predictions of actual organizational behaviour.

While these comparisons establish how alternative financial restrictions affect spending levels, they do not independently determine whether the resulting payroll distributions become more equal. This distinction motivates the subsequent examination of payroll inequality using the Gini coefficient.

Table 13: Impact of Alternative Salary Cap and Payroll Floor Scenarios on MLB Payroll Inequality (2026)

Scenario	Original Gini	Adjusted Gini	Gini Change	Percentage Change
A - MLB Proposal	0.2360	0.0846	-0.1514	-64.16%
B - Restrictive	0.2360	0.0676	-0.1684	-71.35%
C - Flexible	0.2360	0.1823	-0.0537	-22.76%

While the previous analyses examined how alternative salary cap and payroll floor systems influence total payroll expenditures, they did not directly measure how these financial restrictions affect the distribution of spending across MLB organizations. To investigate this relationship, the Gini coefficient was introduced as a measure of payroll inequality.

The Gini coefficient is commonly used in economics to quantify inequality within a distribution, with values ranging from zero, representing perfect equality, toward one, representing extreme inequality. In this analysis, the coefficient was calculated using the payroll figures of all 30 MLB organizations for the 2026 season.

The original payroll distribution produced a Gini coefficient of 0.2360, providing a baseline against which the three hypothetical scenarios could be compared. Adjusted payroll distributions were then constructed by reducing payrolls above each scenario's salary cap to the maximum permitted amount and increasing payrolls below the floor to the required minimum. Teams already within the permitted range retained their original payroll values.

The resulting Gini coefficients demonstrated reductions in payroll inequality under all three scenarios.

Scenario A, based on MLB's proposed 258.2 million dollar cap and 180.2 million dollar floor, reduced the Gini coefficient from 0.2360 to 0.0846, representing a 64.16% reduction in measured inequality. Notably, this scenario also produced a net increase of approximately 263.72 million dollars in leaguewide payroll spending.

Scenario B, which imposed a more restrictive 200 million dollar cap and 150 million dollar floor, produced the largest reduction in inequality. Its adjusted Gini coefficient of 0.0676 represented a 71.35% decrease relative to the original distribution. However, this greater level of equality was accompanied by a substantial net reduction of approximately 755.57 million dollars in leaguewide payroll expenditure.

Conversely, Scenario C, which permitted spending between 125 million dollars and 300 million dollars, produced a smaller reduction in inequality. Its adjusted Gini coefficient of 0.1823 represented a 22.76% decrease, reflecting the greater variation in payroll expenditure permitted under the wider spending boundaries.

These findings demonstrate that the degree of payroll equality produced by salary regulations depends substantially on the thresholds selected. Furthermore, the comparison between Scenarios A and B illustrates that greater payroll equality does not necessarily correspond with higher aggregate player compensation. A restrictive system may produce a more equal distribution of spending while simultaneously reducing the overall amount spent by MLB organizations.

Nevertheless, reduced payroll inequality should not be interpreted as evidence of improved competitive balance. Although financial restrictions may narrow differences in team spending, the relationship between payroll and regular-season performance established earlier in this study suggests that financial resources alone do not account for the majority of variation in organizational success.

Consequently, the Gini coefficient provides a useful measure of how alternative financial structures influence payroll distributions, but it cannot independently determine their effects on competitive outcomes.

Economic Interpretation and Limitations of Salary Cap and Payroll Floor Simulations

The preceding analyses demonstrate that the financial implications of introducing a salary cap and payroll floor depend substantially on the specific thresholds established. Although all three scenarios reduced measured payroll inequality, their effects on total leaguewide expenditures differed considerably.

Notably, Scenario A, based on MLB's reported salary cap-and-floor proposal, produced a net increase in payroll expenditure of approximately 263.72 million dollars while reducing the Gini coefficient by 64.16%. In contrast, Scenario B generated the greatest reduction in payroll inequality, at 71.35%, but also produced a net decrease of approximately 755.57 million dollars in total payroll expenditure. These findings illustrate that greater equality in the distribution of financial resources does not necessarily correspond with an increase in aggregate player compensation.

The results also highlight the importance of distinguishing between payroll equality and competitive balance. A more equal distribution of payroll expenditure does not inherently guarantee greater equality in regular-season performance. As demonstrated in the earlier sections of this study, team payroll explained approximately 11.68% of the variation in regular-season wins across the 2021–2026 period, while year-over-year changes in payroll exhibited very weak linear relationships with changes in team performance. Therefore, although the proposed financial restrictions would reduce variation in team spending under the simulated conditions, their effect on competitive outcomes cannot be determined from these calculations alone.

Several methodological limitations must also be acknowledged. First, the simulations assume that every organization adjusts its payroll precisely to the nearest permitted boundary, without considering how teams might alter their spending strategies in response to new regulations. In practice, organizations may face existing contractual obligations, roster management constraints, and differences in financial resources that influence their ability to comply.

Second, the analysis applies proposed financial thresholds to payroll data from the 2026 season, rather than modelling the future economic conditions under which such regulations might be implemented. Changes in

league revenues, player salaries, inflation, and organizational finances could substantially affect the number of teams exceeding a salary cap or falling below a payroll floor in subsequent seasons.

Third, MLB's proposed system uses Competitive Balance Tax payroll accounting, which may differ from the payroll figures employed in this study. Consequently, the financial adjustments estimated here should be interpreted as illustrative rather than direct reproductions of MLB's official calculations.

Finally, the simulations do not model the potential effects of salary regulations on player movement, free-agent negotiations, contract structures, or long-term organizational behaviour. These factors could influence how financial restrictions ultimately affect the distribution of talent and competitive outcomes throughout the league.

Despite these limitations, the analysis provides insight into the economic trade-offs associated with alternative salary cap-and-floor structures. The results demonstrate that the specific design of financial regulations is consequential: different combinations of spending ceilings and minimum requirements can produce substantially different outcomes for total payroll expenditure and payroll inequality.

Ultimately, the findings suggest that policies aimed at reducing disparities in organizational spending must be evaluated separately from policies intended to improve competitive balance. While the simulations demonstrate how different financial structures could alter payroll distributions, determining whether such changes would produce a more competitive league requires additional analysis of roster construction, talent distribution, and team performance.